ELEC60029 Semiconductor Devices: *Answers:*

Q1

(a)

(i) In forward bias, diffusion current dominates as charge carriers are injected across the depletion region. Furthermore, there is minimal electric field outside of depletion region. The slope of the minority carrier distributions on either side, and diffusion constant, $D_{n,p}$, for electron/holes on the opposite side determine the current. Recombination means that the minority carriers are progressively lost as they progress to the contact, with the number dependent on the recombination rate, or its reciprocal diffusion time, $\tau_{n,p}$. Majority carriers flow in from the contacts to supply recombination.

[4]

(ii) The average distance a carrier travels before it recombines, the diffusion length, is given by $L = \sqrt{(D\ \tau)}$. The long diode approximation is the limit when the distance to the contact is much longer than L, and the short diode approximation is when the distance to the contact is much shorter than L.

[4]

(b)(i)

$$\rho(x) = e\left(-n + p + N_D^+ - N_A^-\right) = 0$$

Where $\rho(x)$ is the net charge density, n, p are the electron, hole carrier concentrations, and N_D^+, N_A^- are the ionised doner, acceptor concentrations.

[4]

(ii)

$$-n + p + N_D^+ - N_A^- = 0 \quad \text{Charge neutrality}$$

Take n-Si $N_D^+ >> N_A^-$

$$\begin{cases} -n + p + N_D^+ = 0 \\ n \times p = n_i^2 \quad \text{Law of mass action} \end{cases}$$

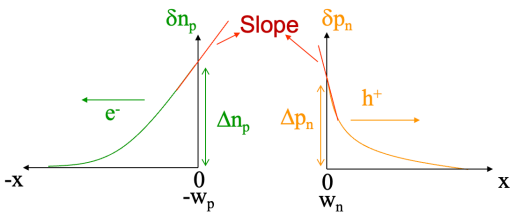
$$\Rightarrow \quad n = \frac{N_D}{2} + \frac{N_D}{2}\sqrt{1 + \frac{4n_i^2}{N_D^2}}$$

Assume complete ionisation $N_D^+ = N_D$

[6]

(c) (i)

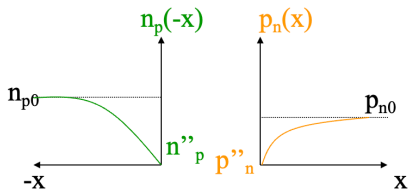
For forward bias:



[2]

ANSWERS

For reverse bias:



[2]

(ii)

For forward bias, (1st diagram above), to find the excess hole (Δp_n) and electron (Δn_p) concentrations, and using the formula sheet, we have:

$n_p' = n_{p0}.\exp(V/V_T) = (n_i^2/N_A)\exp(V/V_T) = ((1.45\times10^{10})^2/10^{16}).\exp(0.5/0.026)$

$= (21025).(0.225\times10^9) = 4.73\times10^{12}$ /cm^3.

Therefore, the excess minority electron concentration $\Delta n_p = n_p' - n_{p0} \sim 4.73\times10^{12}$ /cm^3

[2]

$p_n' = p_{n0}.\exp(V/V_T) = (n_i^2/N_D)\exp(V/V_T) = ((1.45\times10^{10})^2/10^{15}).\exp(0.5/0.026)$

$= (210.25\times10^3).(0.225\times10^9) = 4.73\times10^{13}$ /cm^3.

Therefore, the excess minority hole concentration $\Delta p_n = p_n' - p_{n0} \sim 4.73\times10^{13}$ /cm^3

[2]

For reverse bias (2nd diagram above), we have:

$n_p'' = n_{p0}.\exp(-V/V_T) = (n_i^2/N_A)\exp(-V/V_T) = ((1.45\times10^{10})^2/10^{16}).\exp(-0.5/0.026)$

$= (21025).(4.45\times10^{-9}) = 93.5\times10^{-6} \sim 0.$

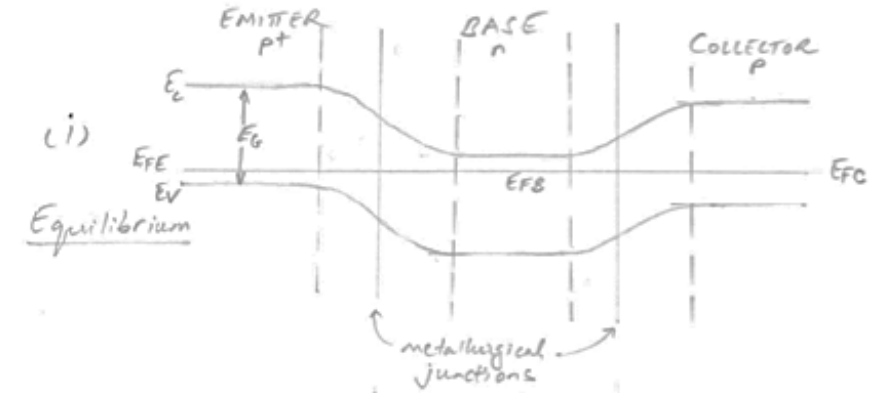
Therefore, the excess minority electron concentration $\Delta n_p = n_p'' - n_{p0} = -n_{p0} = -21\times10^3$ /cm^3, where the minus sign indicates a deficit of carriers from the equilibrium concentration.

Similarly, the excess minority hole concentration $\Delta p_n = p_n'' - p_{n0} = -p_{n0} = -210\times10^3$ /cm^3

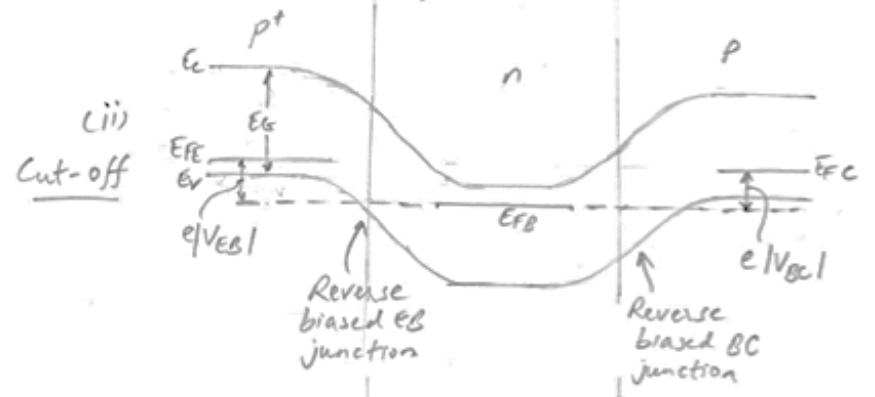
[2]

ANSWERS

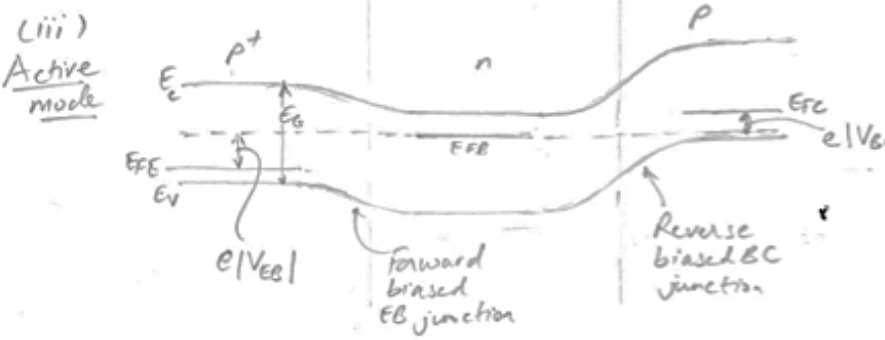
(d)


EMITTER
p+
BASE
n
COLLECTOR
p
(i)
Equilibrium
E_c
E_G
E_{FE}
E_V
E_{FB}
E_{FC}
metallurgical junctions

[4]


(ii)
Cut-off
p^+
n
p
E_c
E_G
E_{FE}
E_V
E_{FB}
E_{FC}
$e|V_{EB}|$
$e|V_{BC}|$
Reverse biased EB junction
Reverse biased BC junction

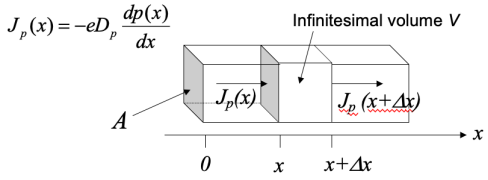
[4]


(iii)
Active mode
p^+
n
p
E_c
E_G
E_{FE}
E_V
E_{FB}
E_{FC}
$e|V_{BC}|$
$e|V_{EB}|$
Forward biased EB junction
Reverse biased BC junction

[4]

Q2

(a) The 1D semiconductor wire is shown in the diagram below. The excess hole concentration $\delta p(x)$ then leads to a diffusion current $J_p(x)$ as shown:



$$\left.\frac{dp}{dt}\right|_{x\to x+\Delta x} = \frac{1}{q}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x}$$

Any generation & recombination processes are neglected in **short** semiconductors.

Rate of variation of the hole concentration = Variation of hole concentration in volume Δx x A/sec

Here, we assume carrier generation/recombination processes can be neglected.

[4 marks for defining this correctly]

This gives:

$$\left.\frac{dp}{dt}\right|_{x\to x+\Delta x} = \frac{1}{q}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x}$$

$$\left.\frac{dp}{dt}\right|_{x\to x+\Delta x} = -\frac{1}{q}\frac{J_p(x)-J_p(x+\Delta x)}{x-(x+\Delta x)}$$

$$\lim_{\Delta x\to 0}: \quad \frac{dp(x,t)}{dt} = -\frac{1}{q}\frac{dJ_p}{dx}$$

[2 marks]

Hence:

$$J_p(x) = -eD_p\frac{dp(x)}{dx}$$

$$\frac{dp}{dt} = \frac{-1}{q}\frac{dJ_p}{dx} = D_p\frac{d^2p}{dx^2}$$

$p = p_o + \delta p$ = equilibrium + excess concentration

$$\frac{d\delta p}{dt} = D_p\frac{d^2\delta p}{dx^2}$$

[4 marks]

ANSWERS

(b)

i) No external bias thus no net current

$$\left.\begin{array}{l} J_p = 0 \\ J_n = 0 = e\mu_n n(x) E(x) - e D_n \dfrac{dn(x)}{dx} = 0 \end{array}\right\}$$

$$\Rightarrow E(x) = \frac{-eD_n}{e\mu_n n(x)} \frac{dn(x)}{dx} = -\frac{D_e}{\mu_n n(x)} \frac{dn(x)}{dx}$$

introduce $n(x) = N_D(x) = N_0 \exp(-ax)$ & $\dfrac{D}{\mu} = \dfrac{kT}{|q|}$

$$\Rightarrow E(x) = -\frac{kT}{|q|} \frac{N_0 (-a) \exp(-ax)}{N_0 \exp(-ax)} = \frac{kT}{|q|}\,\mathbf{a}$$

[Marks: 3]

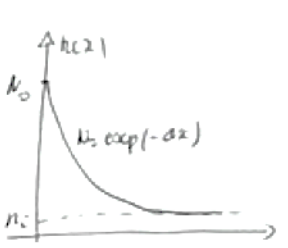
The internal electric field caused by the inhomogeneous doping is

$$E(x) = \frac{kT}{|q|}\mathbf{a} = 0.026\,\mathbf{a}\left(\frac{V}{m}\right)$$

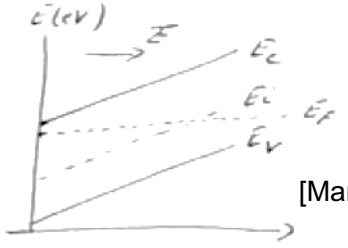
[Marks: 3]

$E(x)$ = 0.052 V/μm

ii)



carrier density variation



potential energy variation.

[Marks: 2]

Interpretation of graphs: e^- diffuse in +x direction due to carrier gradient (left graph). Since there is no net e^- current an internal electric field is built-up (right graph) to counter-balance the diffusion of e^- with a drift of the same number of e^- in the opposite direction.

Since diffusion $\longrightarrow e^-$ $+x$

$\Rightarrow$ drift $\longleftarrow e^-$ $-x$

$\Rightarrow$ e^- roll down the hill by drift

OR E points towards increasing potential energy

$\Rightarrow$ right graph band bending.

[Marks: 4]

Note: important for pn junction.

(c)

For this p^+np BJT, the collector current is given by a hole current, injected across the base by the emitter-base voltage V_{EB}.

$$I_c = I_p$$

[Marks: 2]

The collector current is then given by:

$$I_c = eD_p \left(\frac{p_n' - p_n''}{W_b}\right) A = e\frac{kT}{e}\mu_p \frac{p_n'}{W_b} A$$

[Marks: 2]

Where we have neglected p_n''. Including the effect of V_{EB}:

$$I_c = A\frac{kT}{e}\mu_p \frac{p_{n0}}{W_b} exp\left(\frac{eV_{EB}}{kT}\right)$$

[Marks: 2]

Substituting values gives us I_c = 17 mA/cm^2.

[Marks: 2]

ANSWERS

Q3:

(a)

Shockley-Read-Hall recombination, via defect states in the bandgap. [2]

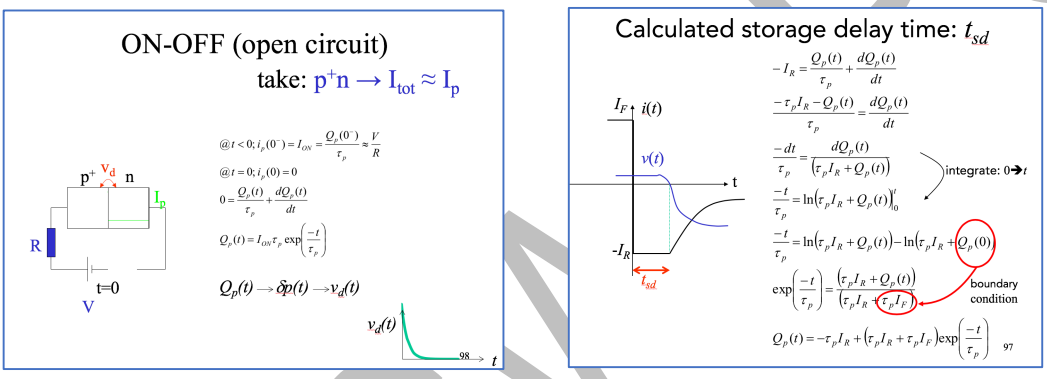
1) Does not require a direct band gap which significantly reduces direct optical recombination [1]
2) Is more likely than thermal excitation across full bandgap due to exponential dependence on the energy difference [1]
3) Requires no electric field to provide additional kinetic energy [1]
4) Does not rely on higher energy states above the conduction band edge that are only likely under heavy doping, such as Auger. [1]

(b)

(i)

Solutions to the given expression, for case A and case B are shown below:

Case A *Case B*



[Give 4 marsk for each of the two solutions, Case A and B]

If $I_F = I_R = I$, then for Case A,

$$Q_1(t) = I\tau_p e^{-\frac{t}{\tau_p}}$$

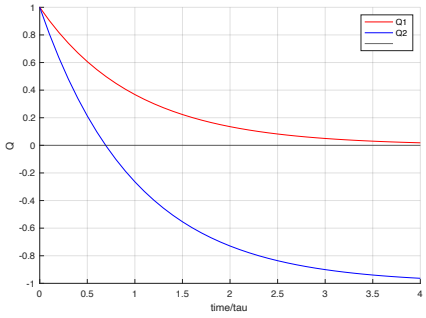
[1]

For Case B,

$$Q_2(t) = I\tau_p(2e^{-\frac{t}{\tau_p}} - 1)$$

[1]

These equations give the following plot (the answer should show the general trend):



[2]

ANSWERS

(ii)

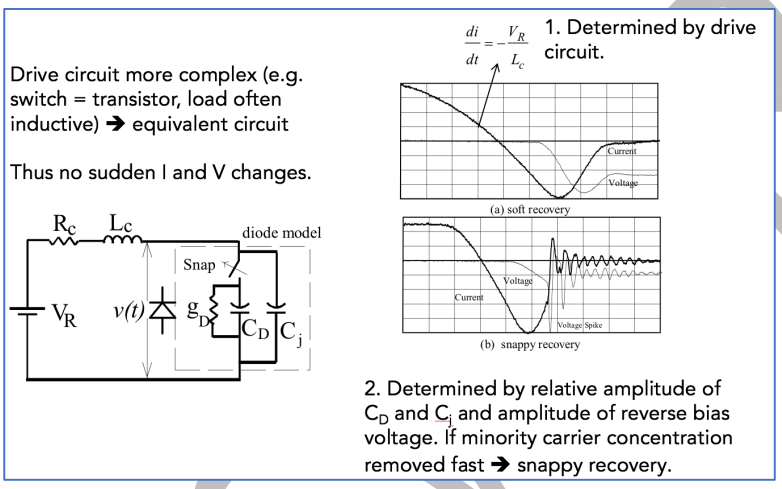
When $Q_2(t) = 0$, $e^{-\frac{t}{\tau_p}} = \frac{1}{2}$, giving $\frac{t}{\tau_p} = \ln 2$. [3]

Substituting into $Q_1(t) = I\tau_p e^{-\frac{t}{\tau_p}}$ gives $Q_1(t) = \frac{I\tau_p}{2}$, which is half the value of $Q_1(0)$. [3]

(c)

Additional delay can occur due to the junction capacitance, this prevents the current returning to zero as soon as the excess charge is removed. This capacitance is due to the charge within the depletion region. With a load resistor, this produces an RC time constant that determines how fast the current returns to zero. [4]

Examples, from notes:



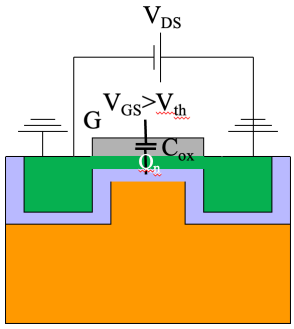
[2]

ANSWERS

Q4:

(a)

For an expression for the inversion layer charge Q_n as a function of $V(x)$, consider the MOSFET cross-section below:



[2]

For the channel along x-axis, $x = 0$ at the source end of the channel region, $V_{DS} = 0$, and applying $V_{GS} \geq V_{th}$, the inversion layer charge:

Q_n = charge/unit area $\Rightarrow$ $Q_n = C_{ox} V$, where V is the net voltage across $C_{ox} = e_0 e_{ox}/t_{ox}$

[2]

We now assume that at $V_{GS} = V_{th}$, $Q_n = 0$

This implies that:

$$Q_n = -C_{ox} (V_{GS} - V_{th})$$

[2]

We now take $V_{DS} > 0$, $V_{GS} \geq V_{th}$

Not only do we now have a voltage in the y direction due to the gate voltage, but also along x due to V_{ds}.

Taking the voltage in the channel due to V_{DS} as $V_x(x)$, and the voltage across the oxide for each point x in the channel as $V_y(x)$:

At $x = 0$ (source end of channel), $V_y(0) = V_{GS}$
At $x = L$ (drain end of channel region), $V_y(L) = V_{GD}$

Furthermore, $V_{GD} = V_{GS} - V_{DS}$

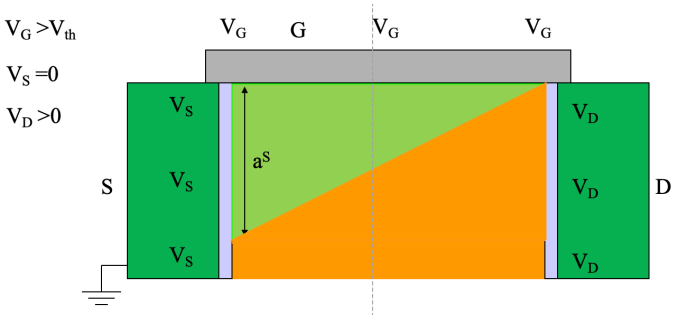
Therefore, using $V_y(x) = V_{GS} - V_x(x)$, the charge in the channel is:

$$Q_n = -C_{ox} (V_{GS} - V_{th} - V_x(x))$$

[4]

ANSWERS

(b) Consider the diagram below, assuming a linear variation in voltage $V(x)$ along the channel, and hence channel depth $a(x)$, as we have only just reached pinch-off. This gives the expressions for Q_n and a at the source and drain as given below:

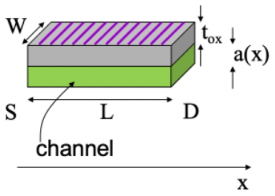


Charge:	$Q_n^S = -C_{ox}(V_{GS} - V_{th})$	$Q_n^D = -C_{ox}((V_{GS} - V_{DS}) - V_{th})$ $Q_n^D = -C_{ox}((V_{GS} - (V_{GS} - V_{th})) - V_{th})$ $Q_n^D = 0 \rightarrow a^D = 0$
Inversion layer width:	$a^S = a_0 Q_n^S$	

[Give 4 marks for each pair of expressions on the source and drain side]

(c) To find the channel drift current I, given the mobile carrier density $n(x)$, electron mobility μ_e, electric field E and channel cross-sectional area $A = Wa(x)$ for the current (see diagram), we have:

$$I = e\, n(x) \mu_e E\, A$$



[4]

Furthermore, $Q_n = -C_{ox}\,(V_{GS} - V_x(x) - V_{th})$ = -e × No. of carriers/gate area(=WL), where the term $V_{GS} - V_x(x)$ is the voltage across the oxide at point x.

Therefore, $n(x)$ can be written as:

$$n(x) = \text{no. of carriers/volume} = Q_n(x)/(-ea(x)) = [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]/ea(x)$$

This gives:

$$\begin{aligned} I &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e E\, A \,/\, a(x) \\ &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e E\, Wa(x) \,/\, a(x) \\ &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e W\, dV_x/dx \end{aligned}$$

[4]

ANSWERS

Integrating:

$$I = C_{ox} (V_{GS} - V(x) - V_{th}) W \mu_e dV_x/dx$$

$$\int_0^L I \, dx = C_{ox} W \mu_e \int_0^{V_{DS}} (V_{GS} - V_x(x) - V_{th}) \, dV_x$$

$$I = C_{ox} \mu_e (W/L) [(V_{GS} - V_{th}) V_{DS} - V_{DS}^2/2]$$

[4]

ANSWERS